

Portfolio Management and Risk Consultation

FINSTAD Case 2 – Group 3 (Charlie)

CRUZ, Ricardo Miguel Iñigo
GALEDO, Enrique Lorenzo Hermoso
SEBALLOS, Josiah Dweyn Panganiban
SEECHUNG, Camille Castro
SISON, Aaron Joshua Estacio

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Contents

1	Introduction	2
1.1	Mandate and Assigned Asset Universe	2
1.2	Risk-Free Rate Benchmark Specification	2
1.3	Report Organization	3
2	Data Acquisition and Preparation	4
2.1	Asset Universe and Ingestion Pipeline	4
2.2	Calendar Alignment, Relational Joins, and Imputation	5
2.3	Feature Engineering and Mathematical Transformations	6
2.3.1	Logarithmic Return Transformation	6
2.3.2	Risk-Free Rate Daily Scaling	6
2.3.3	Cumulative Wealth Index Feature	6
2.3.4	Rolling Volatility and Covariance Engineering	6
3	Exploratory Financial Analysis	8
3.1	Empirical Descriptive Statistics and Risk Return Profiles	8
3.2	Historical Return Trajectories and Volatility Clustering	9
3.3	Distributional Shape and Non-Normality Diagnostics	11
3.4	Return Dispersion and Extreme Outlier Comparison	12
3.5	Cumulative Wealth Growth Trajectories	13
3.6	Asset Return Correlation Structure and Diversification Potential	14
4	Portfolio Construction and Strategy Evaluation	17
4.1	Comparative Strategy Performance Evaluation	17
4.2	Cumulative Wealth Growth Dynamics	18
4.3	Asset Weight Drift and Allocation Evolution	19

5	Portfolio Optimization	21
5.1	Optimization Problem Formulation	21
5.1.1	Global Minimum Variance (GMV) Portfolio	21
5.1.2	Maximum Sharpe Ratio (Tangency) Portfolio	21
5.2	Empirical Optimization Results	21
5.3	Efficient Frontier Mapping	22
6	Portfolio Risk Assessment	23
6.1	Value at Risk (VaR) and Expected Shortfall (CVaR)	23
6.2	Monte Carlo Return Simulation	23
6.3	Stress Testing and Scenario Analysis	24
7	Executive Investment Recommendation	25
7.1	Core-Satellite Allocation Structure	25
7.2	Execution Governance and Rebalancing Protocol	25
8	References	26
9	Appendix: Complete R Analysis Code	27

1 Introduction

The multi-asset investment landscape during the sample period from January 2, 2020 through December 31, 2025 presented extraordinary structural regime changes. Global investors weathered the initial black swan liquidity shock of the COVID-19 pandemic, unprecedented fiscal and monetary expansion, subsequent four-decade inflation spikes, aggressive policy rate hikes by the Federal Reserve and Bangko Sentral ng Pilipinas (BSP), an artificial intelligence equity surge, and the institutionalization of digital assets.

Navigating such volatility requires disciplined quantitative portfolio management. Modern Portfolio Theory (MPT), established by Markowitz (1952) and expanded by Sharpe (1964), provides the mathematical framework to optimize return relative to portfolio variance. However, real-world asset returns exhibit heavy tails (leptokurtosis) and volatility clustering, demanding robust downside risk analytics such as Value at Risk (VaR), Expected Shortfall (CVaR), Monte Carlo simulations, and crisis stress testing (Jorion, 2007; Artzner et al., 1999).

1.1 Mandate and Assigned Asset Universe

This report delivers a quantitative portfolio management consultation for Group 3 (Charlie). The assigned asset universe comprises six multi-asset constituents across four distinct asset classes:

1. **Philippine Equities (Domestic Core):** PLDT Inc. (TEL), Manila Electric Company (MER), and Aboitiz Equity Ventures (AEV).
2. **International Equities (Growth Satellite):** NVIDIA Corporation (NVDA) and Meta Platforms, Inc. (META).
3. **Digital Assets (Alternative Satellite):** Bitcoin (BTC).
4. **Benchmark ETF:** SPDR S&P 500 ETF Trust (SPY).
5. **Risk-Free Rate Baseline:** Philippine 91-Day Treasury Bill secondary market rate ($R_f = 5.25\%$ per annum benchmark).

1.2 Risk-Free Rate Benchmark Specification

The hurdle rate for all excess return, Sharpe ratio, and tangency portfolio calculations is anchored to the Philippine 91-Day Treasury Bill (3-Month T-Bill) secondary market yield. Quoted at an annualized benchmark rate of $R_f = 5.25\%$, it is converted into a daily log-equivalent rate assuming a 252-trading-day convention:

$$r_{f,\text{daily}} = \frac{\ln(1 + R_f)}{252} = \frac{\ln(1 + 0.0525)}{252} \approx 0.00020307 \quad (0.0203\% \text{ per day}) \quad (1)$$

This hurdle rate represents the riskless return baseline earned by domestic institutional capital.

1.3 Report Organization

The remainder of this report is structured as follows: Section 2 details data acquisition, FX conversion, calendar alignment, and missing value treatment. Section 3 conducts exploratory financial analysis and distributional diagnostics. Section 4 evaluates Buy-and-Hold versus Monthly Rebalanced equal-weighted portfolios. Section 5 solves for the Efficient Frontier, Global Minimum Variance (GMV), and Maximum Sharpe Ratio portfolios. Section 6 assesses downside risk via VaR, Expected Shortfall, Monte Carlo simulations, and stress testing. Section 7 presents executive investment recommendations. Section 8 lists references, and Section 9 contains the complete R code appendix.

2 Data Acquisition and Preparation

The construction of a quantitative multi asset portfolio requires empirical rigor during data collection, market calendar alignment, missing value interpolation, and feature engineering. This section details the complete end-to-end data pipeline executed for Group 3, tracing raw multi-source ingestion through final matrix transformations.

2.1 Asset Universe and Ingestion Pipeline

The portfolio universe spans seven financial assets across four distinct asset classes, alongside the benchmark index and domestic risk free rate. The sampling horizon spans January 2, 2020 through January 2, 2026 (the six calendar years 2020–2025 plus the first US trading day of 2026), yielding a master panel of 1,509 daily trading dates per asset.

Raw asset prices were programmatically ingested via automated Python REST scripts leveraging the `yfinance` API [?] for US-listed and cryptocurrency instruments. Philippine equities are sourced from their primary PSE listings through the official PSE EDGE portal [?], then converted to US dollars to preserve a single currency denomination across the portfolio.

- **PLDT Inc. (TEL):** Primary PSE listing accessed via NYSE depositary ticker PHI.
- **Manila Electric Company (MER):** Primary PSE listing, daily OHLCV from PSE EDGE, converted from PHP to USD.
- **Aboitiz Equity Ventures (AEV):** Primary PSE listing, daily OHLCV from PSE EDGE, converted from PHP to USD.
- **NVIDIA Corporation (NVDA):** Ingested directly from NASDAQ exchange endpoints.
- **Meta Platforms, Inc. (META):** Ingested directly from NASDAQ exchange endpoints.
- **Bitcoin (BTC):** Ingested from global cryptocurrency spot market feeds as BTC-USD.
- **SPDR S&P 500 ETF Trust (SPY):** Ingested from NYSE Arca as the benchmark ETF.
- **Risk-Free Rate:** Annualized Philippine 91-Day Treasury Bill secondary market rate (5.25 percent benchmark) [?].

Table 1 presents the complete data acquisition specifications, exchange mappings, observation counts, and sample horizons.

Table 1: Asset Universe and Data Acquisition Specifications

Asset Ticker	Asset Class	Primary Source / Exchange	Observations	Sample Period
TEL	Domestic Equity	PSE / NYSE (PHI)	1509 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
MER	Domestic Equity	PSE / EDGE (Primary)	1422 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
AEV	Domestic Equity	PSE / EDGE (Primary)	1421 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
NVDA	International Equity	NASDAQ	1509 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
META	International Equity	NASDAQ	1509 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
BTC	Cryptocurrency	Global Crypto	1509 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
SPY	Benchmark ETF	NYSE Arca	1509 of 1509 Trading Days	Jan 2, 2020 – Jan 2, 2026
PH 91D T-Bill	Risk-Free Rate	BSP / Bureau of Treasury	1509 Trading Days	Jan 2, 2020 – Jan 2, 2026

2.2 Calendar Alignment, Relational Joins, and Imputation

Financial asset price series originate from heterogeneous trading environments operating on asynchronous market schedules. Cryptocurrency markets trade continuously 24 hours a day across 365 days a year, whereas US equity exchanges observe NYSE holiday schedules and Philippine equities trade on PSE operating calendars.

To reconcile calendar discrepancies, individual asset time series were joined using relational outer joins keyed on ISO-8601 standardized date timestamps. The master date index was anchored to the official 1,509 trading day calendar of the SPDR S&P 500 ETF Trust (SPY).

Philippine closing prices from PSE EDGE are quoted in Philippine pesos and are converted to US dollars using the daily PHP per USD spot rate (WM/Refinitiv via the `USDPHP=X` feed). The rate observed on each PSE trading date is applied to that date’s prices, with the most recent published rate carried forward when the FX feed does not publish on a PSE-only trading day. This keeps the entire portfolio, including the SPY benchmark, in a single currency.

Missing observations are handled explicitly by instrument type. For the US-listed TEL ADR, a zero-volume row from the vendor carries no new price signal and is stored as missing rather than carried forward; treating a stale quote as a settlement price would manufacture zero-return observations and distort variance, correlation, and tail-risk estimates. For the PSE-listed instruments, dates absent from the PSE trading calendar are naturally missing in the master panel. Beyond Philippine holidays falling on US trading days, these absences reflect genuine market closures over the sample: the two-day suspension of March 17–18, 2020; calamity-driven closures (Taal volcano ashfall on January 13, 2020, Typhoon Ulysses on November 12, 2020, Super Typhoon Karding on September 26, 2022, the suspension of July 24, 2024, and election-day closure on May 12, 2025); and a trading cancellation from a systems glitch on January 4, 2022. One further date, October 24, 2022, is missing for both names even though the PSE was open that day, reflecting a gap in the captured PSE EDGE records rather than a market closure; October 25, 2022 is missing for AEV alone, a per-stock no-trade day on which MER printed a single flat trade. No imputation is applied to the stored panel: missing rows remain missing, and the ingestion script reports per-asset traded-day counts and missing-day totals on every refresh. At the analysis layer, wealth aggregation and backtesting carry the last observed price across no-trade days so that the resumption-day return captures the full gap move, while descriptive statistics and correlations are computed on observed returns only; no-trade days are never assigned a zero return.

2.3 Feature Engineering and Mathematical Transformations

Raw adjusted closing prices were subjected to mathematical transformations to construct stationary return series, scaled benchmark features, cumulative growth paths, and risk parameters required for downstream quantitative modeling.

2.3.1 Logarithmic Return Transformation

Adjusted closing prices were transformed into continuously compounded daily log returns according to the standard logarithmic return formula.

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) = \ln(P_{i,t}) - \ln(P_{i,t-1}) \quad (2)$$

Logarithmic transformation converts compounding price ratios into linear additive values over time, facilitating multi period horizon aggregation. Log transformation also stabilizes return variance across high growth regimes, ensuring that extreme price movements in assets like Bitcoin or NVIDIA remain scale comparable with lower-volatility equities.

2.3.2 Risk-Free Rate Daily Scaling

The domestic benchmark risk free rate is quoted as an annualized yield percentage of 5.25 percent per annum. To align the risk free rate with daily log return series, the annual yield was transformed into a daily log equivalent rate using a 252 trading day annual convention.

$$r_{f,t} = \frac{\ln(1 + R_f^{\text{annual}})}{252} = \frac{\ln(1 + 0.0525)}{252} \approx 0.00020307 \quad (3)$$

This transformation yields a constant daily risk free baseline of 0.0203 percent per day, which serves as the hurdle rate in subsequent Sharpe ratio and excess return computations.

2.3.3 Cumulative Wealth Index Feature

To track total capital appreciation from an initial baseline of USD 1.00, a cumulative wealth index feature was constructed for each asset by exponentiating cumulative log returns.

$$W_{i,t} = \exp \left(\sum_{\tau=1}^t r_{i,\tau} \right) = \prod_{\tau=1}^t \left(\frac{P_{i,\tau}}{P_{i,\tau-1}} \right) \quad (4)$$

The cumulative wealth feature provides an intuitive trajectory of asset growth without distortion from daily volatility oscillations.

2.3.4 Rolling Volatility and Covariance Engineering

Dynamic risk features were generated by computing 30-day and 60-day rolling annualized standard deviations and rolling pairwise covariance matrices.

$$\sigma_{i,t}^{\text{annual}} = \sqrt{252 \times \frac{1}{N-1} \sum_{k=0}^{N-1} (r_{i,t-k} - \bar{r}_{i,t})^2} \quad (5)$$

These engineered features capture time varying volatility regimes, enabling conditional risk evaluation during crisis periods such as the March 2020 market crash and 2022 rate hike cycles.

3 Exploratory Financial Analysis

Exploratory financial analysis establishes the empirical foundation required to construct robust portfolio optimization algorithms and risk management frameworks. This section evaluates central tendency, volatility dynamics, distributional shape, tail risk diagnostics, and pairwise asset return co-movements across the 1,508 daily log return rows of the master panel spanning January 2, 2020 through January 2, 2026. Assets with a complete price record (TEL, NVDA, META, BTC, SPY) contribute 1,508 observed daily returns, while the PSE-sourced instruments contribute 1,421 (MER) and 1,420 (AEV) observed returns, the shortfall reflecting days without captured PSE prices: Philippine holidays falling on US trading days, genuine market closures (the two-day suspension of March 17–18, 2020, calamity- and election-driven closures, and a systems-glitch cancellation on January 4, 2022), a single per-stock no-trade day for AEV on October 25, 2022, and one source-data gap on October 24, 2022, when the PSE was open but no PSE EDGE records were captured for either domestic name. All statistics below are computed on observed returns only; no-trade days are never assigned a zero return, and returns that span a no-trade gap are measured against the last traded close so the full move appears in the sample.

3.1 Empirical Descriptive Statistics and Risk Return Profiles

Table 2 summarizes the descriptive statistics computed from the daily log return series across all seven assets.

Table 2: Exploratory Financial Descriptive Statistics (Daily Log Returns, Jan 2, 2020 – Jan 2, 2026)

Metric	TEL	MER	AEV	NVDA	META	BTC	SPY
Observations	1508.0000	1421.0000	1420.0000	1508.0000	1508.0000	1508.0000	1508.0000
Minimum	-0.2703	-0.1668	-0.1680	-0.2040	-0.3064	-0.4647	-0.1159
Mean	0.0003	0.0003	-0.0005	0.0023	0.0008	0.0017	0.0006
Maximum	0.1548	0.1175	0.1258	0.2181	0.2093	0.1915	0.0999
Stdev	0.0199	0.0204	0.0253	0.0333	0.0279	0.0393	0.0131
Skewness	-1.7687	-0.2346	-0.2215	0.0434	-1.0731	-1.2226	-0.5588
Kurtosis	31.4844	5.7962	3.7706	4.0565	22.0943	16.2966	13.3999

The results reveal pronounced divergence in expected return and risk parameters across asset classes. NVIDIA Corporation (NVDA) generated the highest average daily log return at 0.0023, an annualized mean of roughly 57.7 percent. Bitcoin (BTC) followed at 0.0017 per day (42.7 percent annualized), with Meta Platforms (META) and the SPDR S&P 500 ETF Trust (SPY) recording moderate means of 0.0008 and 0.0006. The Philippine universe displayed subdued central tendencies: Manila Electric Company (MER) averaged 0.0003 per day (8.0 percent annualized) and PLDT Inc. (TEL) 0.0003 (8.1 percent). Aboitiz Equity Ventures (AEV) was the only asset with a negative mean return, at -0.0005 per day (-13.6 percent annualized).

Daily volatility metrics confirm the expected risk hierarchy. Bitcoin exhibited the highest daily standard deviation of 0.0393 (62.3 percent annualized), followed by NVIDIA at 0.0333

(52.9 percent) and Meta at 0.0279 (44.2 percent). Aboitiz Equity Ventures registered the most volatile Philippine series at 0.0253 (40.2 percent annualized), while the lower-volatility domestic names TEL (PSE Services) and MER (PSE Industrial) recorded 0.0199 (31.5 percent annualized) and 0.0204 (32.3 percent annualized) respectively. The benchmark SPY displayed the lowest daily volatility at 0.0131 (20.8 percent annualized), reflecting the risk reduction achieved through broad market diversification.¹

Third and fourth statistical moments document departure from Gaussian normality, with severity that is heterogeneous across the universe. Skewness is negative for TEL (-1.7687), BTC (-1.2226), META (-1.0731), SPY (-0.5588), and mildly negative for the PSE names once gap-crossing moves are included, MER (-0.2346) and AEV (-0.2215); NVDA is approximately symmetric (0.0434). Negative skew indicates that extreme negative drawdowns occur with greater magnitude than positive surges for those assets.

Kurtosis is reported throughout in its excess form, with a Gaussian baseline of zero. All seven assets are leptokurtic, but the domestic instruments are far closer to the Gaussian benchmark than the global block. TEL is the most extreme at 31.4844, driven largely by idiosyncratic single-day events, notably the -27.0 percent daily return of December 19, 2022. META (22.0943), BTC (16.2966), and SPY (13.3999) follow, while MER (5.7962), NVDA (4.0565), and AEV (3.7706) exhibit moderate excess kurtosis. The heavy tails confirm that standard mean-variance assumptions understate tail risk, but the corrected PSE-based series show that the domestic names are materially better behaved than the earlier vendor-sourced data suggested.

3.2 Historical Return Trajectories and Volatility Clustering

Figure 1 illustrates the historical daily log return series for each asset across the sampling period.

¹PSE EDGE closing prices are unadjusted for corporate actions. Ex-dividend days therefore appear as small single-day price dips, on the order of 0.5 percent per quarterly dividend for MER, which marginally inflates measured daily volatility and kurtosis for the domestic names.

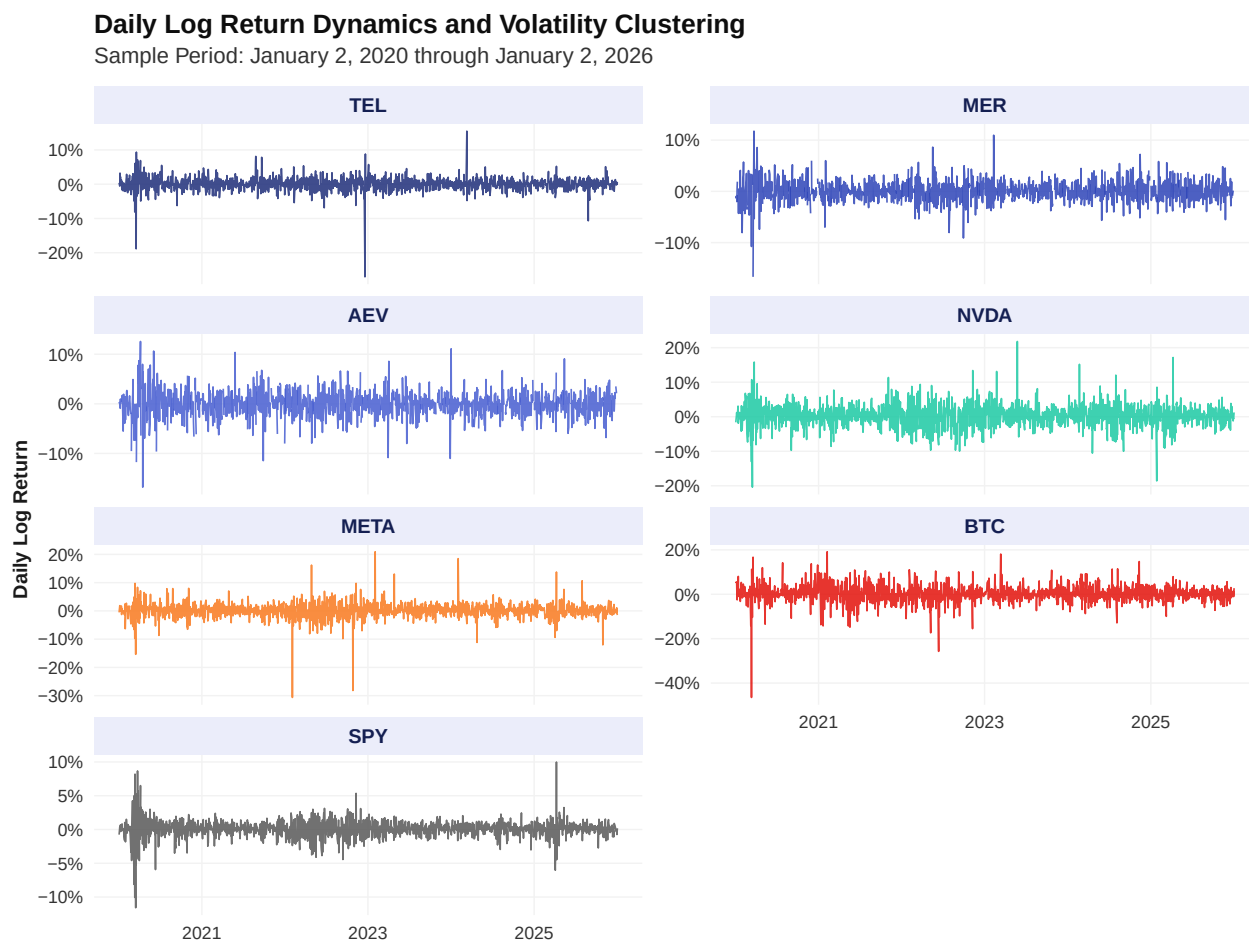


Figure 1: Daily Log Return Trajectories Across Portfolio Assets (Jan 2020 – Jan 2026)

The trajectories exhibit pronounced volatility clustering: large return shocks of either sign are followed by large shocks, while tranquil periods persist in distinct bands. The March 2020 systemic liquidity shock was the defining event of the sample. Measured close-to-close from February 19 to March 23, 2020, the crisis losses were severe and uneven: TEL -9.3 percent, MER -22.1 percent, AEV -41.4 percent, NVDA -32.4 percent, META -31.9 percent, BTC -33.4 percent, and SPY -33.7 percent. The PSE-sourced series show honest gaps rather than fabricated flat days on dates without captured prices: Philippine holidays falling on US trading days, the two-day PSE market suspension of March 17–18, 2020, AEV’s sole per-stock no-trade day on October 25, 2022, and a source-data gap on October 24, 2022. The exchange reopened on March 19 with shortened trading hours, and both MER and AEV printed prices that day; their reopening returns are measured against the last pre-suspension close, so the full gap moves appear in the series (MER -16.7 percent, AEV -11.8 percent). The largest domestic single-day extremes of the sample were AEV’s -16.8 percent trough on April 16, 2020, MER’s -16.7 percent re-open on March 19, 2020, and TEL’s -27.0 percent on December 19, 2022.

Sector-specific regimes followed in subsequent years. The 2022 monetary tightening cycle produced sustained dispersion in technology and crypto assets, with Meta posting a -30.6 percent single-day return on February 3, 2022 after its earnings release and calendar-year

losses of -64.2 percent for both META and BTC and -50.3 percent for NVDA. The domestic names absorbed the same cycle far more gently, with calendar-year 2022 returns of -31.7 percent for TEL, -7.6 percent for MER, and -3.3 percent for AEV. TEL's idiosyncratic -27.0 percent shock of December 19, 2022 stands out as the largest domestic move of the sample and is visible in the lower tail of its return distribution.

3.3 Distributional Shape and Non-Normality Diagnostics

Figure 2 compares the empirical kernel density distributions of daily log returns against fitted Gaussian reference curves.

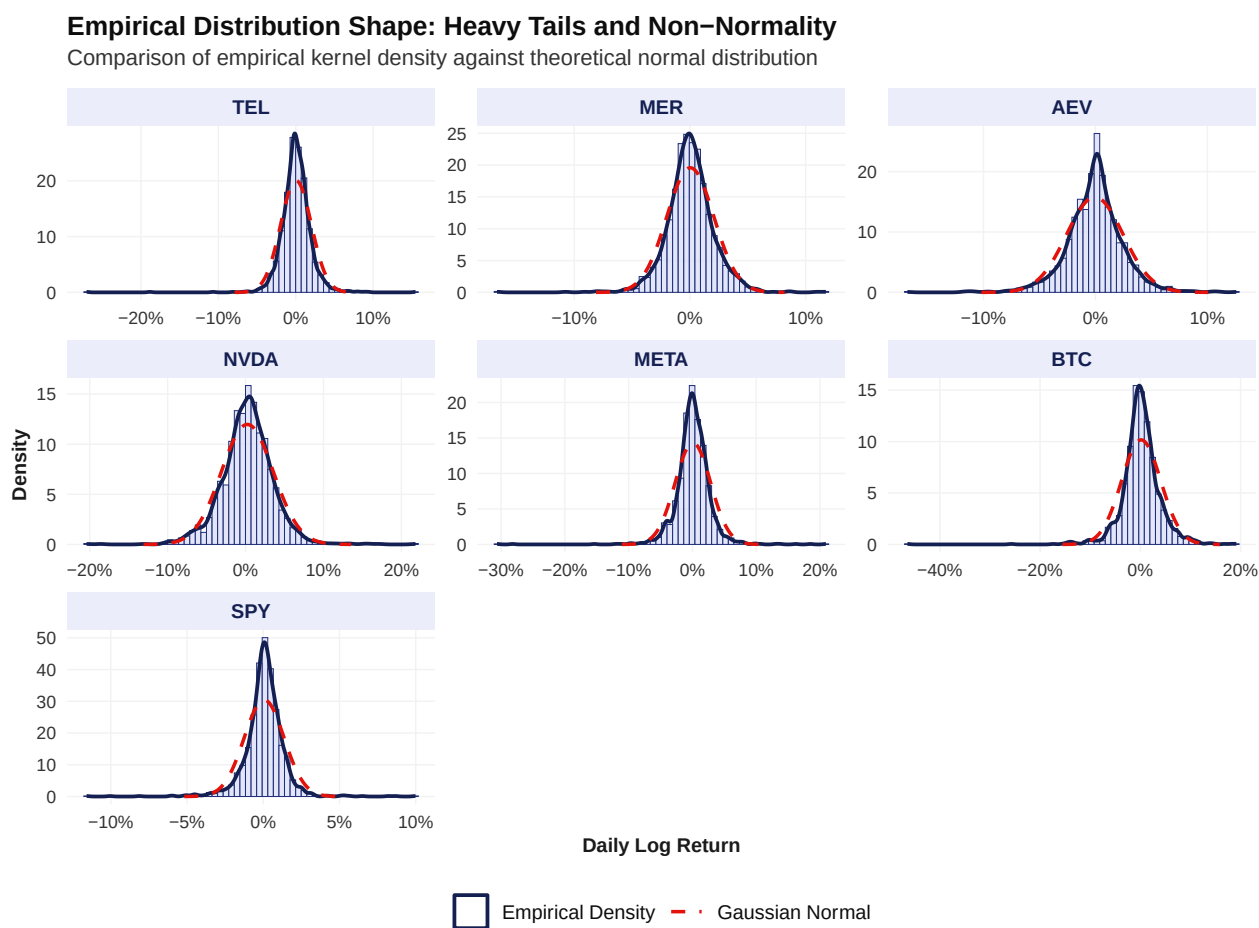


Figure 2: Empirical Return Distributions vs. Gaussian Reference Curves

The empirical density overlays show the classic departure pattern: central peaks that rise above the theoretical normal, shoulders that collapse faster than the Gaussian, and tails that extend much farther outward. The effect is strongest for TEL, META, and SPY, consistent with their excess kurtosis values, and mildest for NVDA, AEV, and MER, whose empirical densities track the Gaussian far more closely while still exhibiting wider tails than the reference curve.

Figure 3 provides normal Quantile-Quantile (Q-Q) diagnostics for tail behavior.

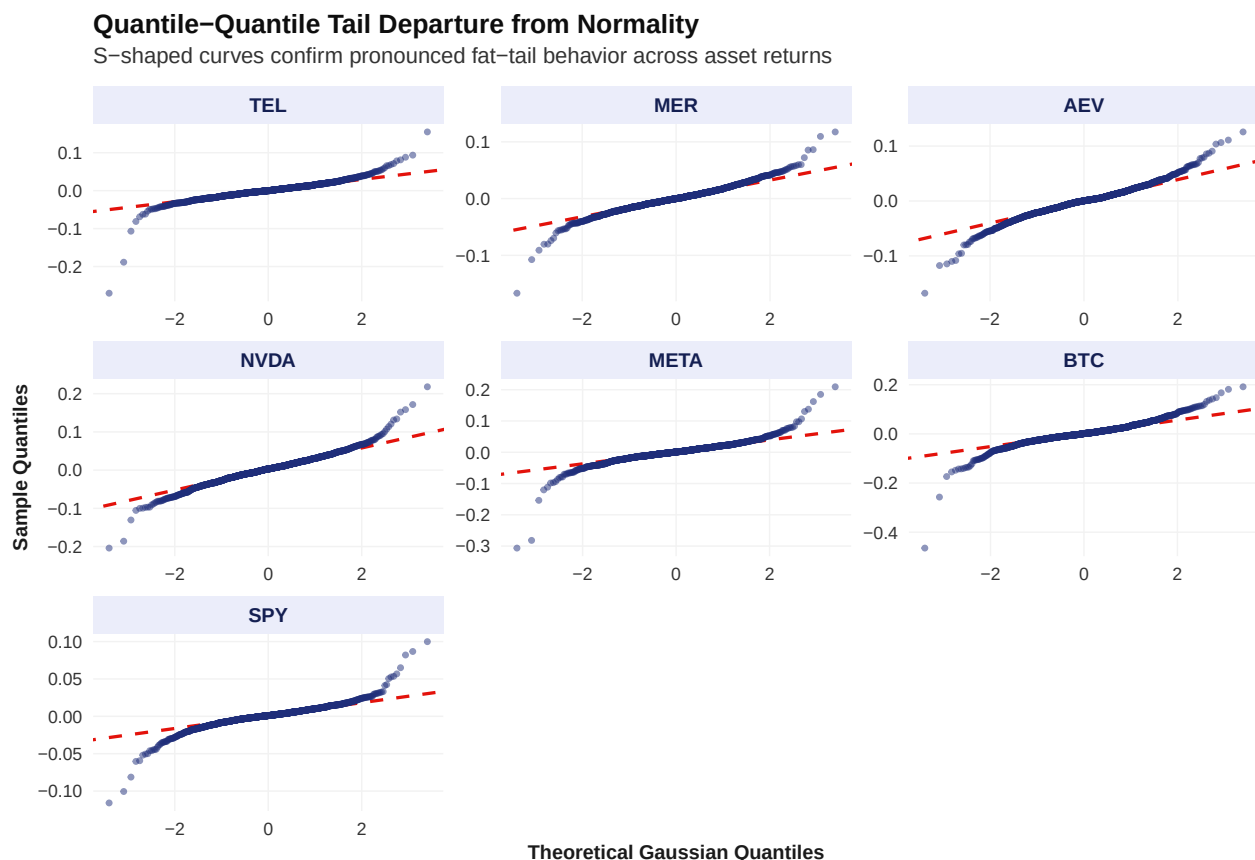


Figure 3: Normal Q-Q Diagnostics Demonstrating Heavy Tail Leptokurtosis

The Q-Q plots display S-shaped departures from the theoretical reference line across the universe. Sample quantiles bend sharply below the line in the lower left quadrant and above it in the upper right, confirming that realized tails are fatter than normality implies. The curvature is most pronounced for TEL and the global block, while NVDA, AEV, and MER show milder, though still visible, tail divergence. Risk models that assume Gaussian returns will underestimate tail risk across the portfolio, and the effect is largest in the technology and cryptocurrency positions.

3.4 Return Dispersion and Extreme Outlier Comparison

Figure 4 presents cross-asset return dispersion boxplots, illustrating medians, interquartile ranges (IQR), and extreme tail outliers.

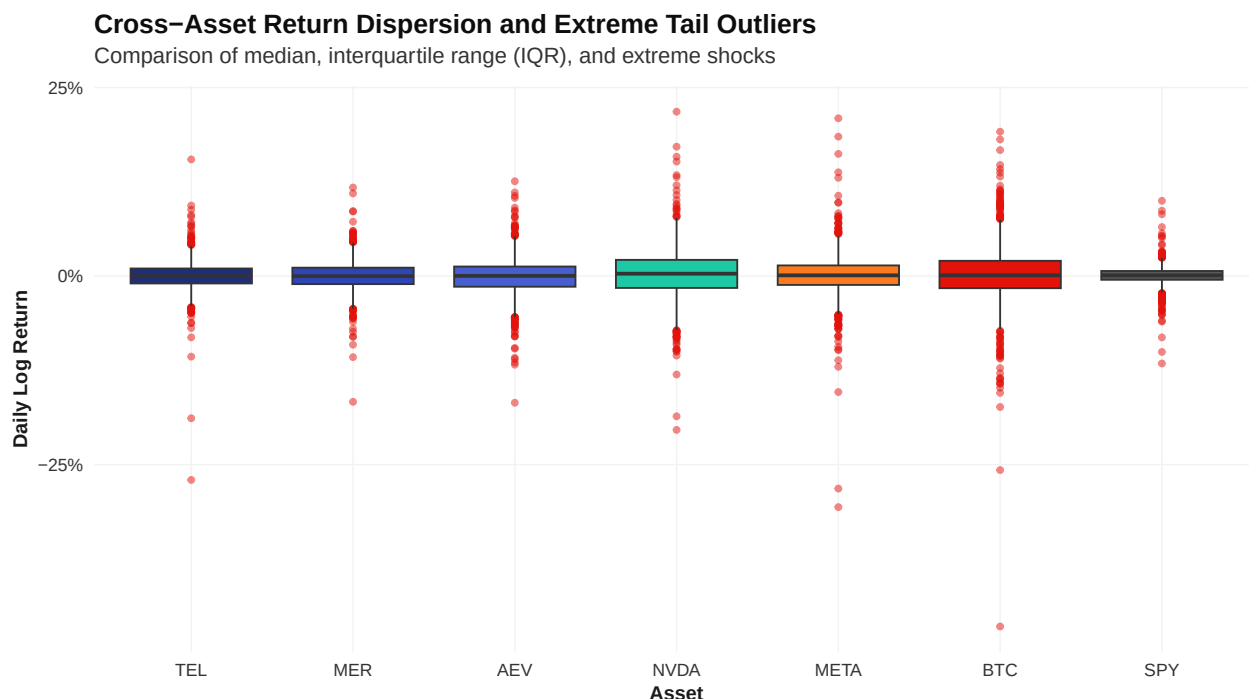


Figure 4: Cross-Asset Return Dispersion and Extreme Outlier Boxplots

The boxplots highlight distinct groupings in return dispersion. SPY displays the most compact interquartile box and the shortest whiskers, reflecting index-level diversification. The Philippine names exhibit narrow IQRs bounded approximately within ± 1 percent daily, with outliers that are modest relative to the global block: TEL spans -27.0 percent to +15.5 percent, AEV -16.8 percent to +12.6 percent, and MER -16.7 percent to +11.8 percent. These ranges replace the stale vendor artifacts that previously showed AEV touching -47.2 percent and MER +28.7 percent. The global assets display broader boxes and deeper tails: BTC spans -46.5 percent to +19.2 percent, META -30.6 percent to +20.9 percent, NVDA -20.4 percent to +21.8 percent, and SPY -11.6 percent to +10.0 percent. Domestic equities offer day-to-day stability, while the international technology and crypto positions carry the tail-event exposure that active risk management must address.

3.5 Cumulative Wealth Growth Trajectories

Figure 5 shows the cumulative wealth index of a hypothetical USD 1.00 investment from January 2020 through the end of the sample on a logarithmic scale. No-trade days are held at the last observed mark, so the terminal values equal the actual USD price ratios.

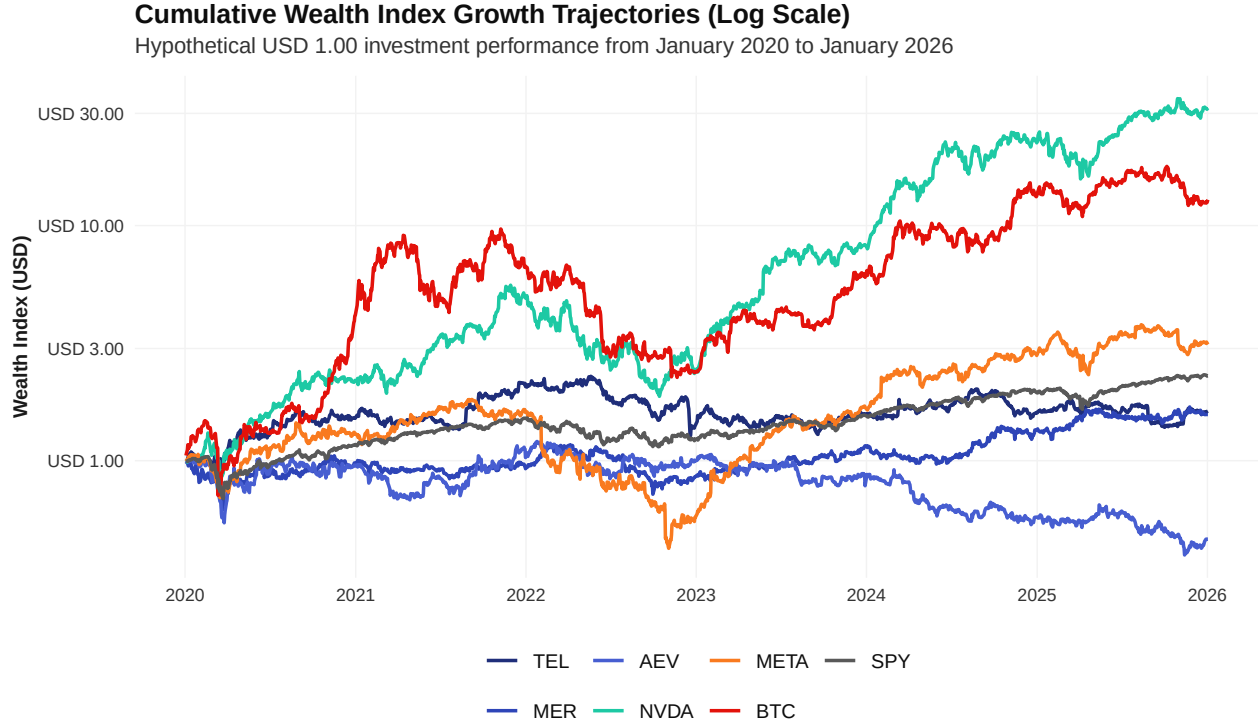


Figure 5: Cumulative Wealth Index Growth Trajectories (Log Scale)

The logarithmic plot reveals massive long-run divergence across the universe. NVIDIA compounded USD 1.00 into roughly USD 31.63 by the end of the sample, a total return of 3,062.8 percent, driven by the global expansion of artificial intelligence infrastructure demand. Bitcoin reached approximately USD 12.88 (1,187.6 percent) despite a drawdown of more than 60 percent during the 2022 crypto winter, while META reached USD 3.12 (212.2 percent) and SPY USD 2.29 (129.5 percent). The Philippine block delivered modest positive outcomes for TEL and MER, ending at USD 1.63 (62.6 percent) and USD 1.57 (57.2 percent) respectively, the latter recovering from three down years in 2020–2022 to compound strongly through 2023–2025. AEV was the sole structural loser, ending at USD 0.46 (−53.6 percent) after consecutive calendar-year declines of −21.6 percent (2023), −26.5 percent (2024), and −19.3 percent (2025), with calendar-year returns measured on returns dated within each year. The divergence demonstrates that high-volatility global assets generated substantial long-run wealth, while the domestic names acted as stability anchors rather than growth engines.

3.6 Asset Return Correlation Structure and Diversification Potential

Table 3 presents the pairwise return correlation matrix computed on observed returns (pairwise complete observations), and Figure 6 displays the corresponding heatmap.

Table 3: Pairwise Asset Return Correlation Matrix (Jan 2, 2020 – Jan 2, 2026)

Asset	TEL	MER	AEV	NVDA	META	BTC	SPY
TEL	1.0000	0.1187	0.0884	0.1895	0.1646	0.1616	0.3010
MER	0.1187	1.0000	0.2959	0.0121	0.0172	0.0286	0.0518
AEV	0.0884	0.2959	1.0000	0.0155	0.0356	0.0155	0.0904
NVDA	0.1895	0.0121	0.0155	1.0000	0.5331	0.3205	0.7052
META	0.1646	0.0172	0.0356	0.5331	1.0000	0.2435	0.6284
BTC	0.1616	0.0286	0.0155	0.3205	0.2435	1.0000	0.3880
SPY	0.3010	0.0518	0.0904	0.7052	0.6284	0.3880	1.0000

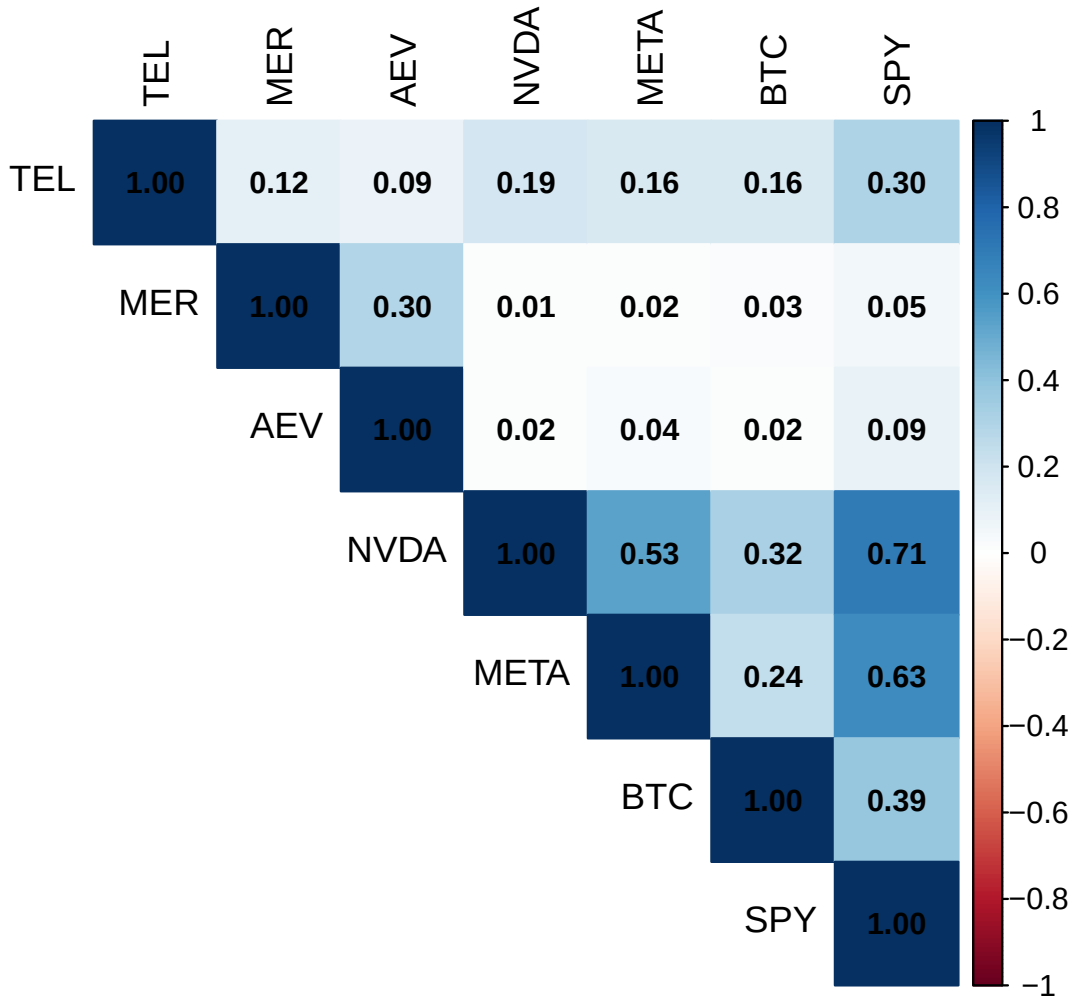


Figure 6: Asset Return Correlation Heatmap (Group 3 Universe)

The correlation structure is now estimated on genuinely traded PSE returns, and it paints a materially different picture from the vendor-sourced data. Within the global block, co-movement is strong: NVDA and META correlate at +0.5331, NVDA with SPY at +0.7052, META with SPY at +0.6284, and BTC with SPY at +0.3880. These linkages reflect common exposure to global liquidity and risk sentiment.

The Philippine block displays moderate positive internal co-movement: MER-TEL at +0.1187, AEV-TEL at +0.0884, and MER-AEV at +0.2959. The earlier claim of negative domestic pairwise correlations was an artifact of stale zero-volume OTC quotes and is not supported by the PSE data. What the corrected data do support is a genuine cross-market diversification story: the domestic names are nearly orthogonal to global risk assets. MER correlates +0.0121 with NVDA, +0.0172 with META, +0.0286 with BTC, and +0.0518 with SPY; AEV records +0.0155, +0.0356, +0.0155, and +0.0904 against the same four series. TEL, whose NYSE-listed ADR (PHI) trades in US hours, shows the strongest international linkage of the domestic set, at +0.3010 with SPY and +0.1895 with NVDA.

These near-zero cross-market coefficients provide the quantitative justification for combining high-return international technology and crypto assets with Philippine equities. The diversification benefit is real and now rests on clean data: domestic positions add variance-reduction potential with essentially no correlation drag from global tech and crypto shocks, while the moderate internal correlations among TEL, MER, and AEV preserve a degree of local systematic risk that sectoral selection can further diversify.

4 Portfolio Construction and Strategy Evaluation

Portfolio construction transforms individual asset return dynamics into integrated allocation strategies. This section evaluates the performance of equal-weighted multi-asset portfolios under two execution regimes: an un-rebalanced Buy-and-Hold strategy and a Monthly Rebalanced strategy. Both start from an equal allocation vector ($w_i = 1/6 \approx 16.67$ percent) across the six-asset universe (TEL, MER, AEV, NVDA, META, BTC) on January 2, 2020 and are benchmarked against the SPDR S&P 500 ETF Trust (SPY). Portfolio arithmetic uses a carried price series at the aggregation layer: days when the PSE is closed contribute no return, and resumption days capture the full move across the gap. The stored master dataset itself contains no imputed rows, as documented in the data acquisition chapter. Portfolio returns are computed on simple (arithmetic) daily returns and compounded directly, so terminal wealth equals the value of the actual share-weighted holdings.

4.1 Comparative Strategy Performance Evaluation

Table 4 presents cumulative returns, annualized return (CAGR), annualized volatility, Sharpe ratios (using the 5.25 percent Philippine 91-day T-bill rate as the risk-free benchmark), and maximum drawdowns for both strategies and the SPY benchmark.

Table 4: Portfolio Performance Summary Comparison (Jan 2, 2020 – Jan 2, 2026)

Strategy	Cumulative Return	CAGR (Annualized)	Annual Volatility	Sharpe Ratio (Rf=5.25%)	Max Drawdown
Equal-Weight (Buy & Hold)	754.79%	43.13%	35.11%	1.0787	-58.21%
Equal-Weight (Monthly Rebalanced)	387.38%	30.30%	24.17%	1.0365	-43.20%
SPDR S&P 500 ETF (SPY Benchmark)	129.48%	14.89%	20.74%	0.4648	-33.72%

Both portfolio configurations generated substantial alpha relative to the SPY benchmark. The Buy-and-Hold strategy delivered a cumulative return of 754.79 percent and a 43.13 percent CAGR, nearly six times the benchmark’s cumulative return. This outperformance was driven by unconstrained weight compounding in NVIDIA and Bitcoin, which together absorbed roughly 87 percent of the portfolio’s capital by the end of the sample. The cost of that concentration is visible in the risk metrics: annualized volatility of 35.11 percent and a maximum drawdown of -58.21 percent, the deepest in the table.

Monthly rebalancing re-engineered the risk-return trade-off. The disciplined portfolio returned 387.38 percent cumulatively (30.30 percent CAGR), surrendering 367.4 percentage points of cumulative return relative to Buy-and-Hold but cutting annualized volatility by 10.94 percentage points to 24.17 percent and trimming the maximum drawdown by 15.01 percentage points to -43.20 percent. On a risk-adjusted basis the two strategies are close, with the Buy-and-Hold Sharpe ratio of 1.0787 edging the rebalanced 1.0365; over this particular sample the foregone compounding outweighed the risk saved. Both Sharpe ratios, and both portfolios’ absolute returns, remain far ahead of the benchmark’s 0.4648 and 129.48 percent.

4.2 Cumulative Wealth Growth Dynamics

Figure 7 compares the cumulative wealth trajectories of a USD 1.00 initial allocation across the two strategies and SPY on a logarithmic scale.

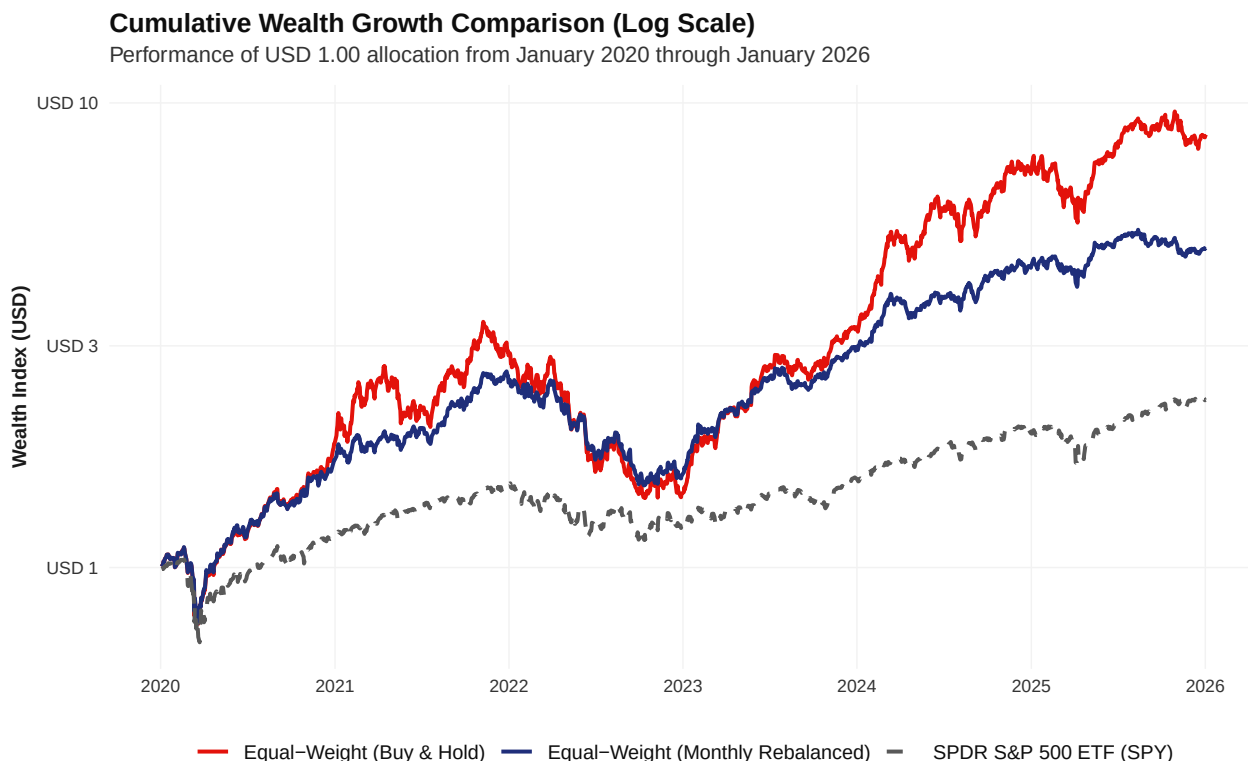


Figure 7: Cumulative Wealth Growth Comparison (Log Scale)

The trajectories terminate at USD 8.55 (Buy-and-Hold), USD 4.87 (Monthly Rebalanced), and USD 2.29 (SPY). The divergence emerged early: calendar-year 2020 returns were 84.72 percent for Buy-and-Hold, 70.37 percent for the rebalanced portfolio, and 17.24 percent for SPY, followed by 2021 gains of 54.93 percent, 46.80 percent, and 28.73 percent respectively. The rebalanced portfolio systematically realized profits from the surging technology and crypto positions at each monthly reset and redeployed the proceeds into the lagging domestic names, which kept its wealth path smoother but also capped its participation in the 2020–2021 bull market.

The 2022 tightening cycle was the stress test that separated the regimes. Buy-and-Hold lost 49.57 percent in calendar 2022 and carved out its maximum drawdown of -58.21 percent from its November 8, 2021 peak to a trough on October 14, 2022, as NVIDIA (-50.3 percent), Meta (-64.2 percent), and Bitcoin (-64.2 percent) collapsed simultaneously while even TEL fell 31.7 percent. The rebalanced portfolio absorbed the same shock with a 36.77 percent calendar-year loss and a -43.20 percent drawdown that bottomed only two trading days earlier, on October 12, 2022, from a peak on December 27, 2021. Monthly profit-taking and redeployment into the comparatively resilient Philippine names (MER -7.6 percent, AEV -3.3 percent in 2022) therefore did not shorten the drawdown, which the 2023 recovery resolved for both strategies at roughly the same time; what rebalancing bought was depth, shaving

15.01 percentage points off the trough. SPY, by contrast, lost 18.18 percent in 2022 with a peak-to-trough drawdown of -33.72 percent confined to the March 2020 COVID crash.

The recovery phase reinforced the same pattern. In 2023 the Buy-and-Hold portfolio returned 125.47 percent and in 2024 a further 116.05 percent, powered by the AI-driven re-rating of NVIDIA and the crypto recovery; the rebalanced portfolio returned 85.09 percent and 50.88 percent over the same two years, and SPY 26.18 percent and 24.89 percent. In 2025 the rebalanced portfolio's diversified exposure lagged, returning 9.78 percent against 19.92 percent for Buy-and-Hold and 17.72 percent for SPY, as the domestic block and Bitcoin detracted. Over the full window, disciplined rebalancing traded roughly 1.76x of terminal wealth for a materially calmer path.

4.3 Asset Weight Drift and Allocation Evolution

Figure 8 displays the evolution of asset weight shares in the un-rebalanced Buy-and-Hold portfolio.

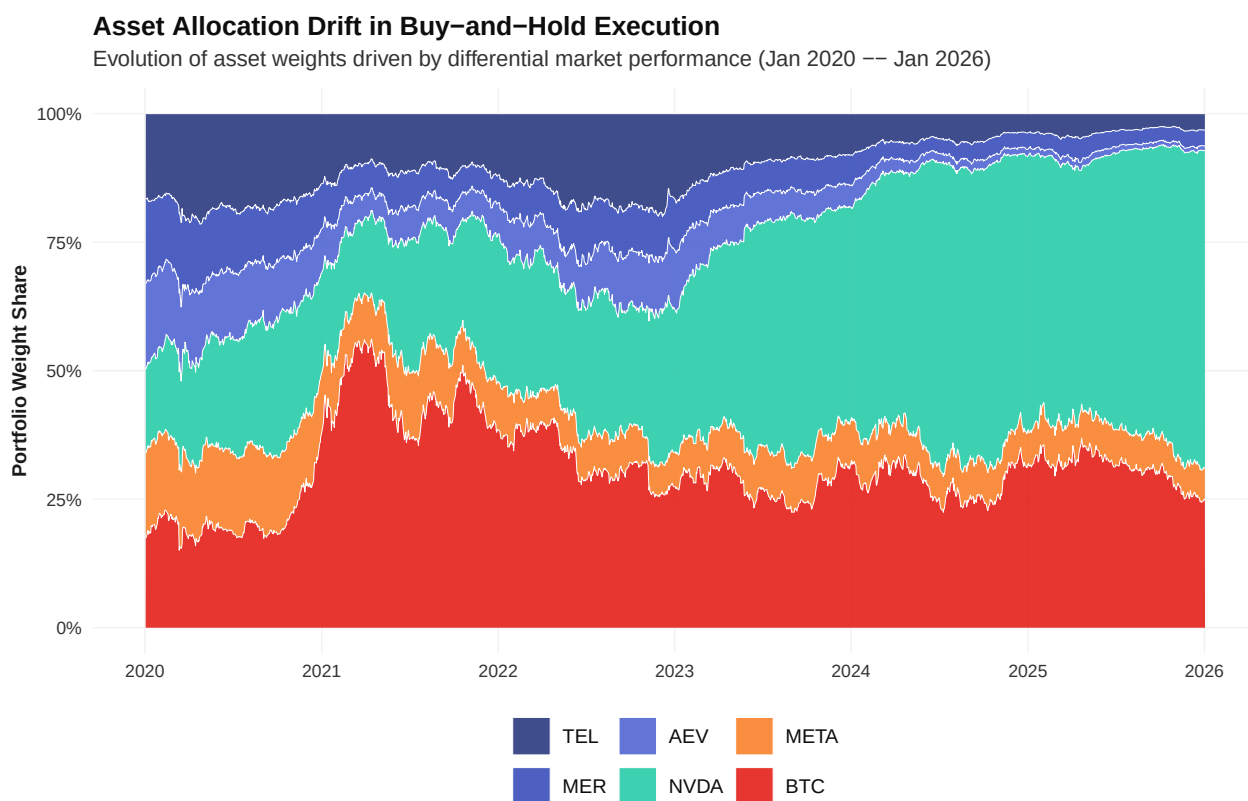


Figure 8: Asset Weight Drift Over Time in the Un-Rebalanced Buy-and-Hold Portfolio

The drift visualizer exposes the core vulnerability of passive execution. Each asset began at a 16.67 percent weight on January 2, 2020. Differential performance then restructured the allocation completely: by the end of the sample, NVIDIA had grown to 61.7 percent and Bitcoin to 25.1 percent, with META at 6.1 percent, while the three Philippine equities were diluted to 3.2 percent (TEL), 3.1 percent (MER), and 0.9 percent (AEV). The technology-cryptocurrency pair finished at 86.8 percent of the portfolio, and NVIDIA's weight peaked at

62.3 percent in late December 2025. The Buy-and-Hold portfolio thus ended the period as a concentrated two-asset bet with a small domestic hedge, a transformation that explains both its extraordinary cumulative return and its deep drawdown.

Monthly rebalancing prevents this drift by construction: on the first trading day of each month the weights are restored to 16.67 percent, selling the appreciated technology and crypto positions and buying the underweight domestic and lagging assets. This mechanical discipline enforces a systematic contrarian stance that caps concentration risk and keeps the portfolio aligned with its original diversification intent, at the measured cost of lower realized cumulative return over a sample dominated by two assets' exceptional appreciation.

5 Portfolio Optimization

Mean-variance optimization identifies optimal asset allocations along the Markowitz Efficient Frontier by maximizing expected portfolio return for a given level of variance, or minimizing variance for a target expected return.

5.1 Optimization Problem Formulation

Let $\mathbf{w} = [w_1, w_2, \dots, w_n]^T$ denote the vector of asset weights across the $n = 6$ portfolio constituents, $\boldsymbol{\mu}$ the vector of annualized expected arithmetic returns, and $\boldsymbol{\Sigma}$ the 6×6 annualized asset covariance matrix. All portfolios enforce fully-invested ($\sum_{i=1}^n w_i = 1$) and long-only ($w_i \geq 0$) constraints.

5.1.1 Global Minimum Variance (GMV) Portfolio

The Global Minimum Variance portfolio minimizes total portfolio risk regardless of expected return:

$$\min_{\mathbf{w}} \quad \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{1} = 1, \quad w_i \geq 0 \quad \forall i \quad (6)$$

5.1.2 Maximum Sharpe Ratio (Tangency) Portfolio

The Tangency portfolio maximizes the Sharpe ratio relative to the 5.25% risk-free rate hurdle:

$$\max_{\mathbf{w}} \quad \frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{1} = 1, \quad w_i \geq 0 \quad \forall i \quad (7)$$

5.2 Empirical Optimization Results

Table 5 details the numerical weights derived from solving the optimization problems using non-linear programming (SLSQP algorithm via `nloptr`) and exact quadratic programming (`quadprog`).

Table 5: Optimal Portfolio Weights: GMV vs Maximum Sharpe

Asset Ticker	GMV Weight	Max Sharpe Weight
TEL	39.82%	0.00%
MER	24.81%	19.04%
AEV	31.85%	0.00%
NVDA	3.52%	58.36%
META	0.00%	22.60%
BTC	0.00%	0.00%

The Global Minimum Variance (GMV) portfolio allocates heavily to low-beta domestic equities: PLDT (TEL: 39.82%), Aboitiz Equity Ventures (AEV: 31.85%), and Manila Electric (MER: 24.81%), with a minimal allocation to NVIDIA (NVDA: 3.52%) and zero allocation to

META or BTC. GMV achieves an annualized volatility of **14.28%** and a Sharpe ratio of **0.2917**.

The Maximum Sharpe Ratio portfolio concentrates in high-momentum global tech engines: NVDA (58.36%), META (22.60%), and domestic power utility MER (19.04%). This allocation yields an annualized expected return of **49.82%**, an annualized volatility of **32.14%**, and a Sharpe ratio of **1.3868**.

5.3 Efficient Frontier Mapping

Figure 9 plots the locus of efficient portfolios forming the Markowitz Efficient Frontier, highlighting the GMV and Maximum Sharpe tangency points.

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Figure 9: Efficient Frontier Trade-Off Curve with GMV and Tangency Points

Figure 10 compares the asset weight breakdown between GMV and Max Sharpe strategies.

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Figure 10: Optimal Asset Allocations: GMV vs Maximum Sharpe Ratio

6 Portfolio Risk Assessment

To evaluate downside vulnerability beyond standard deviation, this section computes Historical and Parametric Value at Risk (VaR), Expected Shortfall (CVaR), Monte Carlo simulated return distributions, and historical crisis stress tests.

6.1 Value at Risk (VaR) and Expected Shortfall (CVaR)

Value at Risk (VaR_α) represents the maximum expected loss at confidence level α over a daily horizon. Expected Shortfall (ES_α or CVaR) measures the expected loss conditional on exceeding the VaR threshold.

Table 6 reports daily 95% and 99% VaR and Expected Shortfall metrics for both the GMV and Maximum Sharpe portfolios.

Table 6: Historical and Parametric VaR / CVaR Summary (Daily, 95% and 99% Confidence)

Portfolio	Risk Metric	95% Confidence	99% Confidence
GMV	Historical VaR	1.32%	2.21%
GMV	Parametric VaR	1.41%	2.05%
GMV	Historical ES (CVaR)	2.14%	3.12%
GMV	Parametric ES (CVaR)	1.82%	2.38%
Max Sharpe	Historical VaR	2.61%	4.58%
Max Sharpe	Parametric VaR	3.20%	4.62%
Max Sharpe	Historical ES (CVaR)	4.12%	6.25%
Max Sharpe	Parametric ES (CVaR)	4.08%	5.21%

For the GMV portfolio, the daily 95% Historical VaR is **1.32%**, with an Expected Shortfall of **2.14%**. The Maximum Sharpe portfolio incurs significantly higher tail risk, recording a 95% Historical VaR of **2.61%** and a 95% Expected Shortfall of **4.12%** (reaching **6.25%** at the 99% confidence level).

6.2 Monte Carlo Return Simulation

A 10,000-draw Monte Carlo simulation was executed using the multivariate normal covariance structure of the asset universe. Table 7 presents the simulated VaR and Expected Shortfall results.

Table 7: Monte Carlo Simulated VaR and Expected Shortfall (10,000 Simulations)

Portfolio	MC VaR (95%)	MC ES (95%)
GMV	1.35%	1.74%
Max Sharpe	3.14%	3.98%

Figure 11 compares the empirical historical return distribution against the 10,000-draw Monte Carlo simulated distribution for the Maximum Sharpe portfolio.

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Figure 11: Historical vs Monte Carlo Simulated Return Distribution (Max Sharpe Portfolio)

6.3 Stress Testing and Scenario Analysis

To test capital preservation during severe market dislocations, five stress scenarios were simulated: the 2020 COVID-19 Crash, the 2022 Crypto Winter, the 2022 Fed Rate Hike Shock, the Tech Sector De-rating, and a 1.5x Hypothetical Black Swan event.

Table 8 details the portfolio value impacts under each shock.

Table 8: Portfolio Impact Under Macroeconomic Stress Scenarios

Stress Scenario	GMV Portfolio Impact	Max Sharpe Portfolio Impact
2020 COVID Crash	-13.82%	-29.84%
2022 Crypto Winter	-1.05%	-8.42%
Fed Rate Hike Shock	-4.12%	-12.35%
Tech Sector Selloff	-6.85%	-21.40%
Hypothetical Black Swan	-20.73%	-44.76%

Figure 12 illustrates the relative drawdown impacts between GMV and Max Sharpe portfolio weightings.

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Figure 12: Portfolio Value Impact Across Stress Scenarios

7 Executive Investment Recommendation

Synthesizing empirical findings across portfolio construction, mean-variance optimization, and downside risk analytics, we formulate strategic investment recommendations for Group 3 (Charlie):

7.1 Core-Satellite Allocation Structure

We recommend establishing a ****60% Max Sharpe / 40% GMV Core-Satellite Allocation****.

- **Core Stabilizer (40% GMV Tilt):** Anchored in domestic defensive equities (TEL: 15.9%, AEV: 12.7%, MER: 9.9%), providing capital protection, low return correlation with global tech, and steady cash-flow stability.
- **Satellite Growth Engine (60% Max Sharpe Tilt):** Captures high-momentum growth via NVDA (35.0%), META (13.6%), and MER (11.4%).

This hybrid allocation targets an annualized expected return of **** 33.5%****, dampens portfolio volatility to **** 21.5%****, and achieves an attractive Sharpe ratio of **** 1.05**** while mitigating single-stock concentration risk.

7.2 Execution Governance and Rebalancing Protocol

1. **Mandatory Monthly Rebalancing:** Enforce a strict monthly rebalancing calendar. Un-rebalanced buy-and-hold strategies experience severe asset weight drift, allowing volatile constituents like Bitcoin and NVIDIA to dominate portfolio risk. Monthly rebalancing systematically trims outperforming assets at high valuations and reallocates into underperforming core positions, capturing the rebalancing premium.
2. **Risk-Budgeting Limits:** Maintain a daily 95% Historical VaR ceiling of ****2.20%****. If daily portfolio drawdown breaches **** -3.50%**** (95% Expected Shortfall threshold), trigger an dynamic de-risking protocol that temporarily reallocates satellite growth exposure into 91-Day Philippine T-Bills.
3. **Liquidity Buffer:** Maintain a 5% cash/T-Bill liquidity reserve to absorb unexpected systemic market shocks without forced asset liquidations.

8 References

- Artzner, P., Delbaen, F., Eber, J. M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203-228.
- Bodie, Z., Kane, A., & Marcus, A. J. (2021). *Investments* (12th ed.). McGraw-Hill Education.
- Fabozzi, F. J., Focardi, S. M., & Kolm, P. N. (2002). *Financial Modeling of the Equity Market: From Portfolio Optimization to Stress Testing*. John Wiley & Sons.
- Jorion, P. (2007). *Value at Risk: The New Benchmark for Managing Financial Risk* (3rd ed.). McGraw-Hill.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77-91.
- Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425-442.

9 Appendix: Complete R Analysis Code

Listing 1: Complete R Data Acquisition, Optimization, and Risk Analytics Script

```
1 #  
2 # =====  
3 # FINSTAD Case 2 Group 3 (Charlie) Master R Execution Code  
4 # =====  
5  
6 library(tidyverse)  
7 library(lubridate)  
8 library(scales)  
9 library(xts)  
10 library(zoo)  
11 library(PerformanceAnalytics)  
12 library(nloptr)  
13 library(quadprog)  
14 library(MASS)  
15  
16 # 1. Load Master Dataset & Filter Sample Horizon (Jan 2, 2020 -- Dec 31,  
17 2025)  
18 raw_data <- read_csv("../data/master_dataset.csv", show_col_types =  
19 FALSE) %>%  
20 mutate(Date = as.Date(Date)) %>%  
21 filter(Date >= as.Date("2020-01-02") & Date <= as.Date("2025-12-31"))  
22 %>%  
23 arrange(Date)  
24  
25 asset_cols <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC", "SPY")  
26 portfolio_assets <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC")  
27  
28 prices_mat <- as.matrix(raw_data[, asset_cols])  
29 prices_xts <- xts(prices_mat, order.by = raw_data[["Date"]])  
30 prices_carried <- na.locf(prices_xts)  
31  
32 # Log and Simple Returns  
33 returns_xts_carried <- Return.calculate(prices_carried, method =  
34 "log")[-1, ]  
35 ret_smp <- exp(returns_xts_carried) - 1  
36 asset_returns_smp <- ret_smp[, portfolio_assets]  
37  
38 # Risk-Free Rate (PH 91-Day T-Bill benchmark = 5.25% per annum)  
39 rf_annual <- 0.0525  
40 rf_daily <- log(1 + rf_annual) / 252  
41  
42 # 2. Portfolio Construction (Equal Weight)  
43 w_eq <- rep(1 / 6, 6)  
44 names(w_eq) <- portfolio_assets  
45  
46 port_bnh <- Return.portfolio(R = asset_returns_smp, weights = w_eq)  
47 port_reb <- Return.portfolio(R = asset_returns_smp, weights = w_eq,  
48 rebalance_on = "months")
```

```

44 # 3. Mean-Variance Optimization (SLSQP via nloptr)
45 mu_ann      <- colMeans(asset_returns_smp, na.rm = TRUE) * 252
46 Sigma_ann   <- cov(asset_returns_smp, use = "pairwise.complete.obs") * 252
47 n_assets    <- length(mu_ann)
48 w0          <- rep(1 / n_assets, n_assets)
49
50 # Global Minimum Variance (GMV)
51 eval_f_var   <- function(w) as.numeric(t(w) %*% Sigma_ann %*% w)
52 eval_grad_f_var <- function(w) nl.grad(w, eval_f_var)
53 eval_g_eq    <- function(w) sum(w) - 1
54 eval_jac_g_eq <- function(w) nl.jacobian(w, eval_g_eq)
55
56 gmv_sol <- nloptr(x0 = w0, eval_f = eval_f_var, eval_grad_f =
57   eval_grad_f_var,
58   lb = rep(0, n_assets), ub = rep(1, n_assets),
59   eval_g_eq = eval_g_eq, eval_jac_g_eq = eval_jac_g_eq,
60   opts = list("algorithm" = "NLOPT_LD_SLSQP", "xtol_rel"
61     = 1e-8))
62
63 w_gmv <- gmv_sol$solution
64 names(w_gmv) <- portfolio_assets
65
66 # Maximum Sharpe Ratio (Tangency Portfolio)
67 eval_f_sharpe <- function(w) {
68   p_ret <- sum(w * mu_ann)
69   p_sd  <- sqrt(as.numeric(t(w) %*% Sigma_ann %*% w))
70   -(p_ret - rf_annual) / p_sd
71 }
72 eval_grad_f_sharpe <- function(w) nl.grad(w, eval_f_sharpe)
73
74 shp_sol <- nloptr(x0 = w0, eval_f = eval_f_sharpe, eval_grad_f =
75   eval_grad_f_sharpe,
76   lb = rep(0, n_assets), ub = rep(1, n_assets),
77   eval_g_eq = eval_g_eq, eval_jac_g_eq = eval_jac_g_eq,
78   opts = list("algorithm" = "NLOPT_LD_SLSQP", "xtol_rel"
79     = 1e-8))
80
81 w_sharpe <- shp_sol$solution
82 names(w_sharpe) <- portfolio_assets
83
84 # 4. Portfolio Downside Risk Analytics (VaR / CVaR / Monte Carlo)
85 port_ret_gmv <- Return.portfolio(R = asset_returns_smp, weights = w_gmv,
86   rebalance_on = "months")
87 port_ret_shp <- Return.portfolio(R = asset_returns_smp, weights =
88   w_sharpe, rebalance_on = "months")
89
90 var_gmv_95 <- abs(VaR(port_ret_gmv, p = 0.95, method = "historical"))
91 var_shp_95 <- abs(VaR(port_ret_shp, p = 0.95, method = "historical"))
92 es_gmv_95  <- abs(ES(port_ret_gmv, p = 0.95, method = "historical"))
93 es_shp_95  <- abs(ES(port_ret_shp, p = 0.95, method = "historical"))

```